

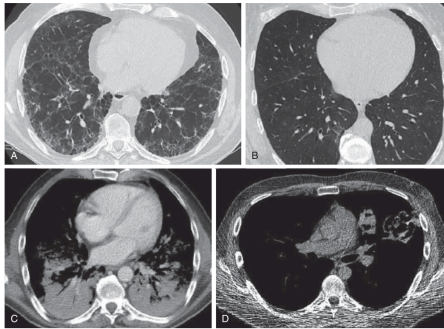


Fig. 32.1 Different patterns of lung involvement encountered in patients with antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis. Computed tomography scan images of various forms of lung injury in patients with ANCA-associated vasculitis. (A) Honeycombing and peripheral reticulation, usual interstitial pneumonitis, as exclusively observed in patients with myeloperoxidase-ANCA vasculitis. (B) Mild ground-glass opacities (diffuse) in a plevascular pattern, here in a patient with myeloperoxidase-ANCA vasculitis. (C) Diffuse alveolar hemorrhage in a patient with proteinase 3-ANCA vasculitis. (D) Two large granulomas in a patient with proteinase 3-ANCA vasculitis.

Fig_32_1

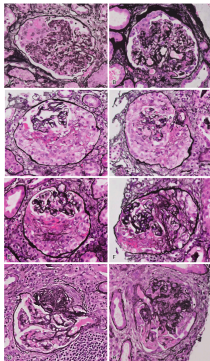


Fig. 32.4 Shapes of a crescent. (A) Hypercellular crescent with fibrinogen. (B) Cellular crescent showing the contribution of crescentic glomerulonephritis to renal function. (C) Crescentic glomerulonephritis with crescentic glomerulonephritis. (D) Crescentic glomerulonephritis with crescentic glomerulonephritis. (E) Crescentic glomerulonephritis with crescentic glomerulonephritis. (F) Crescentic glomerulonephritis with crescentic glomerulonephritis. (G) Crescentic glomerulonephritis with crescentic glomerulonephritis. (H) Crescentic glomerulonephritis with crescentic glomerulonephritis. (I) Crescentic glomerulonephritis with crescentic glomerulonephritis. (J) Crescentic glomerulonephritis with crescentic glomerulonephritis. (K) Crescentic glomerulonephritis with crescentic glomerulonephritis. (L) Crescentic glomerulonephritis with crescentic glomerulonephritis.

Fig_32_4

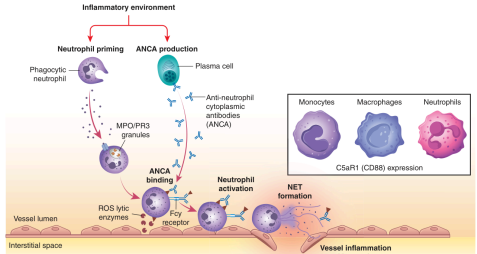


Fig. 32.2 Schematic overview of key pathogenic steps eventually leading to vasculitis. The inflammatory environment in antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis is characterized by immune cell activation, with active contribution of neutrophils, monocytes, and macrophages, which all express the CD34 (CD38). Alternative complement pathway activation, together with other inflammatory stimuli, leads to neutrophil priming and production of ANCA by plasma cells, among others. ANCA bind to neutrophils, which lead to a release of reactive oxygen species and other inflammatory mediators. This leads to activation of the neutrophils and to the formation of neutrophil extracellular traps (NETs), which are key mediators of active vessel inflammation and injury. The role of NETs in antimicrobial defense is well known, but also the role in neutrophil-driven inflammation in diseases such as psoriasis. Autoantibody formation is induced by NETs in ANCA-associated vasculitis, which play a critical role in the onset of autoimmunity. Moreover, key effector cells lead to release of myeloperoxidase and proteinase 3 from granules, which are implicated in disease pathogenesis.

Fig_32_2

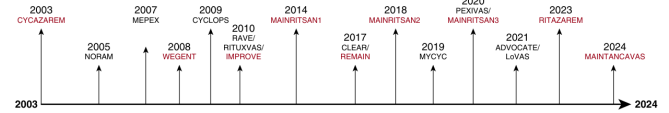


Fig. 32.5 Timeline of relevant therapeutic developments in antineutrophil cytoplasmic antibody-associated vasculitis over the past 2 decades. Trials testing new strategies to induce remission are given in black, while trials testing therapeutic options to maintain remission are highlighted in red.

Fig_32_5

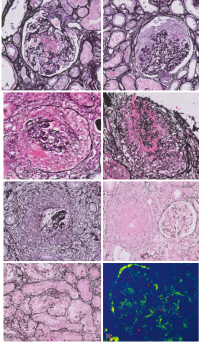


Fig. 32.3 Characteristic lesions in antineutrophil cytoplasmic antibody glomerulonephritis. (A) Crescentic glomerulonephritis with crescentic glomerulonephritis. (B) Crescentic glomerulonephritis with crescentic glomerulonephritis. (C) Crescentic glomerulonephritis with crescentic glomerulonephritis. (D) Crescentic glomerulonephritis with crescentic glomerulonephritis. (E) Crescentic glomerulonephritis with crescentic glomerulonephritis. (F) Crescentic glomerulonephritis with crescentic glomerulonephritis. (G) Crescentic glomerulonephritis with crescentic glomerulonephritis. (H) Crescentic glomerulonephritis with crescentic glomerulonephritis. (I) Crescentic glomerulonephritis with crescentic glomerulonephritis. (J) Crescentic glomerulonephritis with crescentic glomerulonephritis. (K) Crescentic glomerulonephritis with crescentic glomerulonephritis. (L) Crescentic glomerulonephritis with crescentic glomerulonephritis. (M) Crescentic glomerulonephritis with crescentic glomerulonephritis. (N) Crescentic glomerulonephritis with crescentic glomerulonephritis. (O) Crescentic glomerulonephritis with crescentic glomerulonephritis. (P) Crescentic glomerulonephritis with crescentic glomerulonephritis. (Q) Crescentic glomerulonephritis with crescentic glomerulonephritis. (R) Crescentic glomerulonephritis with crescentic glomerulonephritis. (S) Crescentic glomerulonephritis with crescentic glomerulonephritis. (T) Crescentic glomerulonephritis with crescentic glomerulonephritis. (U) Crescentic glomerulonephritis with crescentic glomerulonephritis. (V) Crescentic glomerulonephritis with crescentic glomerulonephritis. (W) Crescentic glomerulonephritis with crescentic glomerulonephritis. (X) Crescentic glomerulonephritis with crescentic glomerulonephritis. (Y) Crescentic glomerulonephritis with crescentic glomerulonephritis. (Z) Crescentic glomerulonephritis with crescentic glomerulonephritis.

Fig_32_3

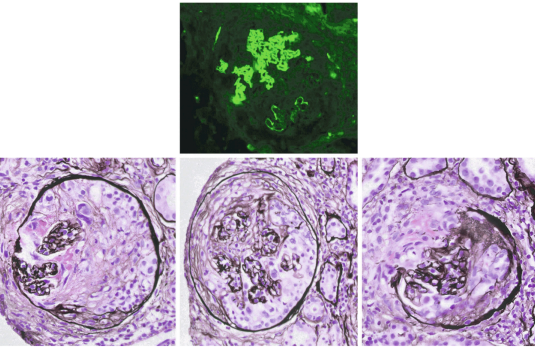


Fig. 32.6 Antiglomerular basement membrane disease: typical linear staining for IgG (top) in combination with crescentic glomerulonephritis (GN). Glomeruli are usually affected in the same stage of crescentic GN. Crescents may be cellular, fibrocellular, or fibrous (lower panel, from left to right).

Fig_32_6

Table 32.1 Overview of Types, Associated Patterns, Immunofluorescence Findings, and Associated Diseases in Presence of "Rapidly Progressive Glomerulonephritis"			
Type	Pattern	Immunofluorescence	Disease(s)
Type 1	Linear	Linear deposits of immunoglobulin G with in situ immune-complex formation	Anti-GBM disease
Type 2	Granular	Granular deposits of immunoglobulins	Lupus nephritis (classes III and IV), IgA vasculitis/IgA nephropathy, postinfectious glomerulonephritis, cryoglobulinemic vasculitis, MPGN (type I and II), fibrillary glomerulonephritis, membranous nephropathy
Type 3	Pauci-immune	"Virtual" absence of glomerular immune deposits	ANCA-glomerulonephritis

ANCA, Antineutrophil cytoplasmic antibody; GBM, glomerular basement membrane; MPGN, membranoproliferative glomerulonephritis.

Table_32_1

Table 32.2 Cluster Analysis of 3868 Patients with Antineutrophil Cytoplasmic Antibody-Associated Vasculitis, Derived from 6 European Registries					
	Severe Kidney	MPO-Kidney	PR3-Kidney	Young Respiratory Cluster	Inflammatory Multisystem Cluster
Patients	555 (14.3%)	782 (20.2%)	683 (17.7%)	646 (16.7%)	1202 (31.1%)
Age	67	63	66	48	55
Follow-up (years)	3	3	4	6	5
Women	44%	50%	36%	59%	50%
MPA	62%	76%	17%	18%	22%
Organs affected					
Constitutional	61%	34%	89%	39%	72%
Musculoskeletal	44%	31%	80%	43%	65%
Skin	7%	12%	42%	21%	30%
ENT	20%	14%	70%	70%	60%
Lung	52%	35%	67%	42%	57%
Cardiovascular	8%	4%	13%	6%	7%
Gastrointestinal	9%	3%	17%	3%	7%
CNS	2%	2%	10%	9%	12%
PNS	9%	4%	25%	15%	28%
Kidney	100%	99%	99%	23%	37%
ANCA	37%	17%	82%	60%	68%
PR3-ANCA	58%	75%	13%	22%	24%
Creatinine (μmol/L)	580; 6.6	240; 2.7	221; 2.5	80; 0.9	78; 0.9
CRP (mg/L)	123	16	110	6	99
Death	31%	21%	23%	6%	15%
KF	42%	28%	20%	2%	4%

Data-driven subclassification revealed the presence of 5 clusters,¹² with highly variable clinical manifestations. Most patients with kidney disease present in clusters associated with acute kidney disease at baseline (severe kidney, myeloperoxidase, and proteinase 3-ANCA clusters). These clusters are associated with higher risks of kidney failure and death. The spectrum of extrarenal disease manifestations and inflammation at baseline is also variable.

ANCA, Antineutrophil cytoplasmic antibody; CNS, central nervous system; CRP, C-reactive protein; ENT, ear, nose, and throat; KF, kidney failure; MPA, microscopic polyangiitis; MPO, myeloperoxidase; PNS, peripheral nervous system; PR3, proteinase 3.

Table_32_2

Table 32.3 Classification Criteria for Granulomatosis with Polyangiitis and Microscopic Polyangiitis as Issued by the American College of Rheumatology and European Alliance of Associations for Rheumatology		
Criteria	2022 ACR/EULAR Classification Criteria for GPA	2022 ACR/EULAR Classification Criteria for MPA
	A total cutoff of ≥5 is needed. Clinical criteria: - Nasal involvement: bloody discharge, ulcers, crusting, congestion, blockage, or septal defect/perforation (+3) - Cartilaginous involvement (inflammation of ear or nose cartilage, hoarse voice or stridor, endobronchial involvement, or saddle nose deformity) (+2) - Conductive or sensorineural hearing loss (+1) - Laboratory, imaging, and biopsy criteria: - Positive test for cANCA or anti-PR3 antibodies (+5) - Pulmonary nodules, mass, or cavitation on chest imaging (+2) - Granuloma, extravascular granulomatous inflammation, or giant cells on biopsy (+2) - Inflammation, consolidation, or effusion of the nasal/paranasal sinuses, or mastoiditis on imaging (+1) - Blood eosinophil count $\geq 1 \times 10^9/L$ (-4) - Positive test for pANCA or anti-MPO antibodies (-1) Patients/Controls (Comparators) Validation Development set: 578/652 Validation set: 146/161 Sensitivity of 92.5% and specificity of 93.8% in the validation cohort (ACR 1990 in same dataset: sensitivity 69.3%, specificity 75.8%)	A total cutoff of ≥5 is needed. Clinical criteria: - Nasal involvement: bloody discharge, ulcers, crusting, congestion, blockage, or septal defect/perforation (+3) - Fibrosis or interstitial lung disease on chest imaging (+3) - Pauci-immune glomerulonephritis on biopsy (+3) - Positive test for cANCA or anti-PR3 antibodies (-1) - Blood eosinophil count $\geq 1 \times 10^9/L$ (-4) Development set: 149/408 Validation set: 142/414 Sensitivity of 90.8% and specificity of 94.2% in the validation cohort (No ACR classification criteria existed for MPA)

ACR, American College of Rheumatology; ANCA, antineutrophil cytoplasmic antibody; EULAR, European Alliance of Associations for Rheumatology; MPA, microscopic polyangiitis; MPO, myeloperoxidase; PR3, proteinase 3.

Table_32_3

Table 32.4 Disease Manifestations in GPA, MPA, PR3-ANCA, and MPO-ANCA Vasculitis				
	GPA	MPA	PR3-ANCA Vasculitis	MPO-ANCA Vasculitis
General	78%	86%	81%	92%
Fever ≥38°C (100.4°F)	31%	40%	44% (≥38.5°C [101.3°F])	
Fatigue	56%	68%	—	
Weight loss ≥2 kg (4.4 lb)	35%	43%	47% (≥3 kg [6.6 lb])	
Arthralgia	55%	32%	56%	
Myalgia	22%	24%	26%	
Cutaneous	35%	30%	34%	17%
Petechiae or purpura	17%	10%	18%	
Mucous membranes/Eyes	38%	13%	28%	10%
Scleritis or episcleritis	14%	1%	5% and 10%	
ENT	82%	26%	81%	2%
Respiratory	63%	63%	68%	52%
Hemoptysis/DAH	21%	19%	18%	22%
Cardiovascular	11%	15%	16%	6%
Abdominal	19%	22%	11%	4%
Renal	59%	62%	58%	79%
Neurologic	31%	37%	30%	39%
Neuropathy	12%	26%	21%	21%
Mononeuritis multiplex	5%	9%	—	
Sensory neuropathy	11%	21%	—	

The frequencies are taken as reported in 674 patients with granulomatosis with polyangiitis (GPA) and 325 patients with microscopic polyangiitis reported in the Diagnostic and Classification Criteria in Vasculitis study,⁹¹ an international study involving 32 countries, and 546 patients with PR3-ANCA vasculitis reported from the French Vasculitis Study Group.⁹² The cohort of patients with MPO-ANCA vasculitis (n = 144) also comprises 6 cases with PR3-ANCA vasculitis and reported data from a single center in Germany.²⁸

ANCA, Antineutrophil cytoplasmic antibody; DAH, diffuse alveolar hemorrhage; DCVAS, Diagnostic and Classification Criteria in Vasculitis; ENT, ear, nose and throat; FVSG, French Vasculitis Study Group; GPA, granulomatosis with polyangiitis; MPA, microscopic polyangiitis; MPO, myeloperoxidase; PR3, proteinase 3.

Table_32_4

Table 32.5 Different Histopathologic Scores have been Reported to Score Biopsies of Patients with ANCA- Glomerulonephritis				
	Berden ⁹³	AKRIS ⁹²	MCCS ⁹³	PACS ⁹⁴
Glomerular lesions	≥50% normal glomeruli (focal) ≥50% cellular crescents (crescentic) ≥50% glomerulosclerosis (sclerotic) All others (mixed)	Normal glomeruli (%) >25 (N0), 10-25 (N1), <10 (N2)	Glomerulosclerosis <10%; 10%-25%; 26%-50%; >50%	% of the following lesions: Cellular crescent Fibrocellular crescent Fibrous crescent Sclerosis (43% predictive of KF)
Tubulointerstitial lesions		IFTA (%) <25 (T0), ≥25 (T1)	Interstitial fibrosis Tubular atrophy (both <10%; 10%-25%; 26%-50%; >50%)	
Vascular lesions			Atherosclerosis None/mild Moderate/severe	
Clinical characteristics		Creatinine (μmol/L) <250 (K0), 250-450 (K1), >450 (K2)		

The ANCA Kidney Risk Score (AKRIS) combines pathology and baseline characteristics and has the best ability to predict kidney failure. The above-mentioned scores have been independently validated, especially the Berden histopathologic classification.

IFTA, interstitial fibrosis, tubular atrophy; KF, kidney failure.

Table_32_5

Table 32.6 Different Steroid Tapering Regimen According to PEXIVAS, LoVAS, and ADVOCATE Trials									
Trial	PEXIVAS			LoVAS		ADVOCATE			
Regimen	Reduced dose	Standard dose		Low dose	High dose				
Weeks	<50/50-75/>75 kg	<50/50-75/>75 kg	Weeks			Weeks	>55 kg	<55 kg	
1-2	25/30/40 mg	50/60/75 mg	1-2	0.5 mg/kg	1.0 mg/kg	1	60 mg	45 mg	
3-4	20/25/30 mg	40/50/60 mg	3-4	0.25 mg/kg	0.8 mg/kg	2	45 mg	45 mg	
5-6	15/20/25 mg	30/40/50 mg	5-6	7.5 mg	0.7 mg/kg	3	30 mg	30 mg	
7-8	12/15/20 mg	25/30/40 mg	7-8	5 mg	0.5 mg/kg	4-6	25 mg	25 mg	
9-10	10/12/15 mg	20/25/30 mg	9-10	4 mg	0.4 mg/kg	7-8	20 mg	20 mg	
11-12	7/10/12 mg	15/20/25 mg	11-12	3 mg	0.35 mg/kg	9-10	15 mg	15 mg	
13-14	6/7/10 mg	12/15/20 mg	13-16	2 mg	15 mg	11-14	10 mg	10 mg	
15-16	5/5/7 mg	10/10/15 mg	17-20	1 mg	12.5 mg	15-20	5 mg	5 mg	
17-18	5/5/7 mg	10/10/15 mg	21-24	0 mg	10 mg	≥21	0 mg	0 mg	
19-20	5/5/5 mg	7/7/7 mg							
21-22	5/5/5 mg	7/7/7 mg							
23-52	5 mg	5 mg							
>52	Investigators' local practice	Investigators' local practice							

The PEXIVAS trial did not mandate steroid withdrawal, and further reduction steps beyond 52 weeks of follow-up were based on the investigators' local practice. The LoVAS trial assessed its primary endpoint at 6 months of follow-up, while steroid withdrawal formed the basis of both primary endpoints of ADVOCATE.

Table_32_6

Table 32.7 Summary of Studies Reporting on Combined use of Cyclophosphamide and Rituximab				
Trial/Study	RITUXIVAS ¹⁹²	Pepper et al., ¹⁹³	Gulati et al., ¹⁹⁴	Cortazar et al., ¹⁹⁵
Patients	33 and 11 (RCT)	69	61	129
Induction	PLEX (6 and 3 patients), RTX arm (4-times 375 mg/m ² and 2 pulses CVC, 15 mg/kg with adjustment, CVC arm according to CYCLOPPS), followed by azathioprine maintenance	RTX (2-times 1000 mg), CVC (6 pulses IV, 500 mg each or 500-750 mg)	PLEX (7 sessions), RTX (6-times 1000 mg), CVC (IV, 10 mg/kg; max, 750 mg first 2 pulses, 4-times 500 mg subsequent)	PLEX (40 patients) RTX (2-times 1000 mg), CVC (3mg, adjusted)
Glucocorticoids	1 g methylprednisolone (4-optimal doses), oral prednisolone down to 5 mg at 4 weeks	1-2 pulse methylprednisolone 350 mg/1 g, oral prednisolone for 1-5 weeks	2.6 g (6 months, initial dosage max. 60 mg)	Pulse methylprednisolone (optional), oral prednisolone taper over 7 months
Maintenance	CYC arm only (AZA)	AZA, first 1mL, MMF, MTX, RTX	MMF or AZA	RTX (500 mg every 4 months)
Serum creatinine (μmol/L)	—	—	558	—
eGFR (mL/min/1.73 m ²)	20 and 12	28.9 and 28.3 (2 groups)	9	18.6 (75 patients with PR3/ANCA)
Remission (at 6 months)	76% and 82% (sustained remission)	86% (47 patients)	94% (60 patients)	84% (109 patients)
Relapse (at 6 months)	86% (38 and 27 mL/min/1.73 m ² at month 12)	Serum creatinine 109 μmol/L and 113 μmol/L at month 12	67% (of 30 with RRT at baseline), of non-RRT patients, eGFR improved to 45 mL/min/1.73 of at 3 years	eGFR 28.9 mL/min/1.73 at baseline, of non-RRT patients, eGFR improved to 45 mL/min/1.73 of at 3 years
Mortality	6 and 2 (18% each)	3 (6.1%) of which 1 was an infectious complication	9 (14.7%), of which 5 were infectious complications (1 COVID-19)	4 (3.1%)

*CYCLOPPS¹⁹¹: 15 mg/kg body weight cyclophosphamide weekly for the first 3 infusions, then 3 further infusions 3 weeks apart, potentially extension to 10 infusions in case of sufficient response. Maximum dose per pulse 1.5 g, adjust for age (60-70 years <2.0 mg/kg, >70 years 2-3 mg/kg), adjust for kidney function (300-500 μmol/L 4-5.7 mg/kg), <2.0 mg/kg), and adjust for potential leukocyte nadir, leukocyte nadir <2.5 × 10⁹/L, <20% and nadir <1.2 × 10⁹/L, <40%.

¹⁹²To control the values for creatinine (μmol/L) or leukocyte (mg/dL), divide by 0.01.

The RITUXIVAS trial was a randomized controlled trial (1:1 randomization), while the others were retrospective observational studies with the limitations inherent to this study design.

AZA, azathioprine; CVC, corticosteroid; CYC, cyclophosphamide; eGFR, estimated glomerular filtration rate; IV, intravenous; MMF, mycophenolate mofetil; MTX, methotrexate; RRT, kidney replacement therapy; PLEX, plasma exchange; PR3, proteinase 3; RCT, randomized controlled trial; RITUX, rituximab; progressive glomerulonephritis; RTX, rituximab.

Table_32_7